ISYE 6740 - Spring 2025
Final Project Report

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Project Title: Leveraging Municipal Codes and Social Media Discourse for Informed Policymaking

Problem Statement

Local governments must understand public opinion to ensure that policy decisions align with the needs and preferences of the communities they serve. Consider the following situation: a local government has an upcoming legislative session during which the city council will discuss updating municipal codes related to a variety of issues such as housing, transportation, and land use. Council members want to know what the public thinks about these issues, specifically what residents want to see changed versus what is working well. They know that residents rarely come in person to city council sessions and are much more likely to express their opinions online. However, the local government has a tight budget and does not have the resources to manually search for and organize social media posts in a way that could inform policy.

This project aims to bridge the gap between local governments and their residents by collecting social media posts which correspond to specific topics outlined in the municipal codes of two selected cities, Denver and Boston, and analyzing them to gauge public opinion on these topics. More specifically, we scraped the text of each city's municipal codes from Municode, extracted the main topics from the text, found Reddit posts that are relevant to a selection of the topics, and determined whether posts contain negative, neutral, or positive sentiments about the topic. We highlight our findings using visualization to show which topics are associated with the most negative or positive posts, thus pinpointing the issues that the public is most concerned about. These findings could inform policy decisions during the council's legislative sessions, supplementing the statements of residents who attend the in-person meetings, to determine the issues which are highest priority for community residents.

Background

This project builds on past studies that have used natural language processing to interpret both local government regulations and public opinion about policy. Studies analyzing local government regulations have extracted text from online news articles (Stacy et al.) as well as digital municipal codes (Bartik et al.; Mleczko and Desmond; Shanks), while analyses of public comments typically draw from government websites (Kowalski et al; Dokshin; Hagen; Kwon et al; Ha and Grubert). The methods employed in these studies range from unsupervised topic modeling (e.g., Latent Dirichlet Allocation) (Shanks; Kowalski et al; Dokshin; Hagen et al) to supervised classification techniques, such as using support vector machines (Kwon et al.) and convolutional neural networks (Ha and Grubert) to categorize public comments by topic and sentiment. While some recent studies use large language models to answer specific questions (Bartik et al.), others use more traditional natural language processing techniques (Stacy et al; Mleczko and Desmond; Shanks). To evaluate their results, the authors typically compare machine learning predictions with data that is manually coded, usually by a team of subject experts (Stacy et al.; Bartik et al; Mleczko and Desmond; Kwon et al; Ha and Grubert). This project combines elements from these studies, drawing from both types of data sources – municipal codes and public comments – and employing both topic modeling and sentiment analysis techniques.

Methodology

In this project, we combine municipal code data from Municode and public comments from Reddit to identify issues in a community where policy is misaligned with public opinion. We employ a combination of data collection via web scraping and API, topic modeling, sentiment analysis, and visualization to achieve this goal.

Data Collection

This project draws from two data sources: Municode and Reddit. Data was collected from a sample of two cities, Denver and Boston, which were selected because of the breadth of their municipal codes and the active Reddit communities in each city.

Municode is a platform that publishes the legal codes and ordinances of municipalities across the United States. The data is organized by state and city or township, each of which maintains its own set of ordinances. The ordinances are composed of a number of articles, with each article divided into a number of sections. The subject and content of the articles and sections vary across municipalities. We scraped the ordinance text for Denver and Boston from Municode using the Selenium library in Python. We saved the text for each city in a JSON file formatted as a list of dictionaries, with each dictionary corresponding to a distinct article from the city's ordinances. As part of our data cleaning effort in preparation of topic modeling, we further split each article text by section, such that each section of an article was its own document. After this division, we obtained 4013 documents for the city of Denver and 3484 documents for the city of Boston.

Reddit, a popular social media platform, hosts topic-specific forums known as subreddits. It has become commonplace for communities, especially larger towns and cities, to have active subreddits in which residents engage in discussion, ask questions, and air grievances specific to their community. Reddit provides an API that enables users to query posts from specific subreddits, along with associated metadata such as the post's date, author, number of upvotes, number of downvotes, and number of comments in JSON format. The Reddit API also allows users to specify query terms which filter responses to only posts deemed most relevant by Reddit's internal relevance-ranking algorithm. While the exact mechanics of this algorithm are not publicly disclosed, it is likely influenced by factors such as keyword matching, user engagement, and recency. We leveraged this functionality to retrieve the top 50 most relevant posts for each topic-phrase identified through LDA topic modeling, along with the associated metadata for each post. We selected 50 posts per topic as the threshold, as we observed a noticeable decline in post quality and relevance beyond that number. After removing posts which contained only photos or videos and no text data, we totaled 559 posts across 12 topics for Denver, and 506 posts across 12 topic for Boston, bringing the total size of our dataset to 1065 posts. The process for determining the topic-phrases used to query the Reddit API will be discussed in more detail in the following section.

Topic Modeling

The topic modeling portion of this project serves two distinct functions: first, to identify patterns in the municipal ordinances which give rise to new topic groupings, and second, to identify the keywords most strongly associated with those topics. This is a necessary step in the analysis process as the ordinance subsections address specific topics, ranging from broad categories (e.g. General Provisions) to highly specific ones (e.g. Massage Businesses). While these groupings are useful for legal categorization, their organization may not align with the way a casual resident of the city would naturally group issues, nor do they necessarily reflect the topics of most interest to the average resident. Through the topic modeling process, we extracted a specific set of topics each associated with 10 most relevant keywords.

We conducted topic modeling using Latent Dirichlet Allocation (LDA), a generative probabilistic model used for unsupervised learning on text datasets. To prepare the data for modeling, we first cleaned the text by making it lowercase and removing unnecessary phrases, punctuation, and stop words. We then lemmatized each word to get its root form and kept only nouns, adjectives, verbs, and adverbs, as these parts of speech are most indicative of a text's meaning. We also formed bigrams and trigrams, which are

sequences of two and three words, respectively, that have a distinct meaning. For example, "real_estate" is a bigram and "performing_arts_complex" is a trigram which were extracted from the texts during pre-processing.

Following preprocessing — which removed footnotes as well as documents consisting solely of noninformative words or characters — we were left with a final set of 3022 documents for Denver and 3139 for Boston (down from 4013 and 3484, respectively). We next tokenized and embedded the documents using the bag-of-words method over the full vocabulary present within the documents, implemented via Gensim's doc2bow function. We also filtered out the extremes of the vocabulary, excluding words which occur less than 15 times across the corpus, or in more than 20% of documents. We then fed the documents into the LDA model. LDA assigns words and documents to topics through an iterative process which uses Gibbs sampling to converge on topic-document distributions and topic-word distributions. Topic-document distributions represent the probability of a document belonging to each topic. Topic-word distributions represent the probability of a word in the corpus belonging to each topic. The model converges when stable distributions are reached. At convergence time, it is assumed that words and documents belong to their highest probability topic. The formula for Gibbs sampling is below (IBM).

$$P(z_i = t \mid z^{-i}, w) = \frac{n_{m,t}^{-i} + \alpha}{\sum_{t'=1}^{T}(n_{m,t'}^{-i} + \alpha)} \times \frac{n_{t,w_i}^{-i} + \beta}{\sum_{v'=1}^{V}(n_{t,v'}^{-i} + \beta)}$$

$P(z_i = t \mid z^{-i}, w)$ represents the probability of assigning word w to topic t, given the current topic assignments of all other words in the corpus. The first term represents the topic-document assignment by calculating the proportion of words assigned to topic t within document d, compared to the total number of words assigned to all topics in the same document. The second term represents the topic-word distribution by calculating the proportion of occurrences of word w_i within topic t, compared to the total number of occurrences of all words in the vocabulary(IBM).

We selected the optimal LDA model by tuning the number of topics T, as well as model hyper-parameters alpha and beta, using C_v coherence score. Alpha modifies the topic-document term, controlling whether documents are likely to have a few dominant topics or a similar probability of many different topics. Beta modifies the topic-word term similarly, controlling whether topics will have a few dominant words or a more diverse set of words. C_v coherence score measures the internal consistency of each topic using information about the co-occurrence patterns of the topic's keywords. A higher C_v coherence score is better. More information on the coherence evaluation criteria is provided in the evaluation section.

The range of topics tested for tuning was between 20 and 50, as this range appeared to offer the best balance between the size of the dataset and the desired level of specificity for each topic. The optimal number of topics determined for Denver was 35 while for Boston the optimal number of topics was 30. After determining the number of topics for each municipality, alpha and beta were tuned by calculating the C_v coherence score for each combination of parameters. The possible values for alpha were symmetric, asymmetric, 0.01, 0.1, and 0.5, while the options for beta included symmetric, 0.01, 0.1, and 0.5. The optimal combination for Denver was an alpha value of 0.01 and a beta value of 0.1, while for Boston it was alpha equal to 0.1 and beta equal to 0.01. The results of the LDA hyperparameter tuning are shown in Figure 1, where the first row corresponds to Denver and the second row corresponds to Boston

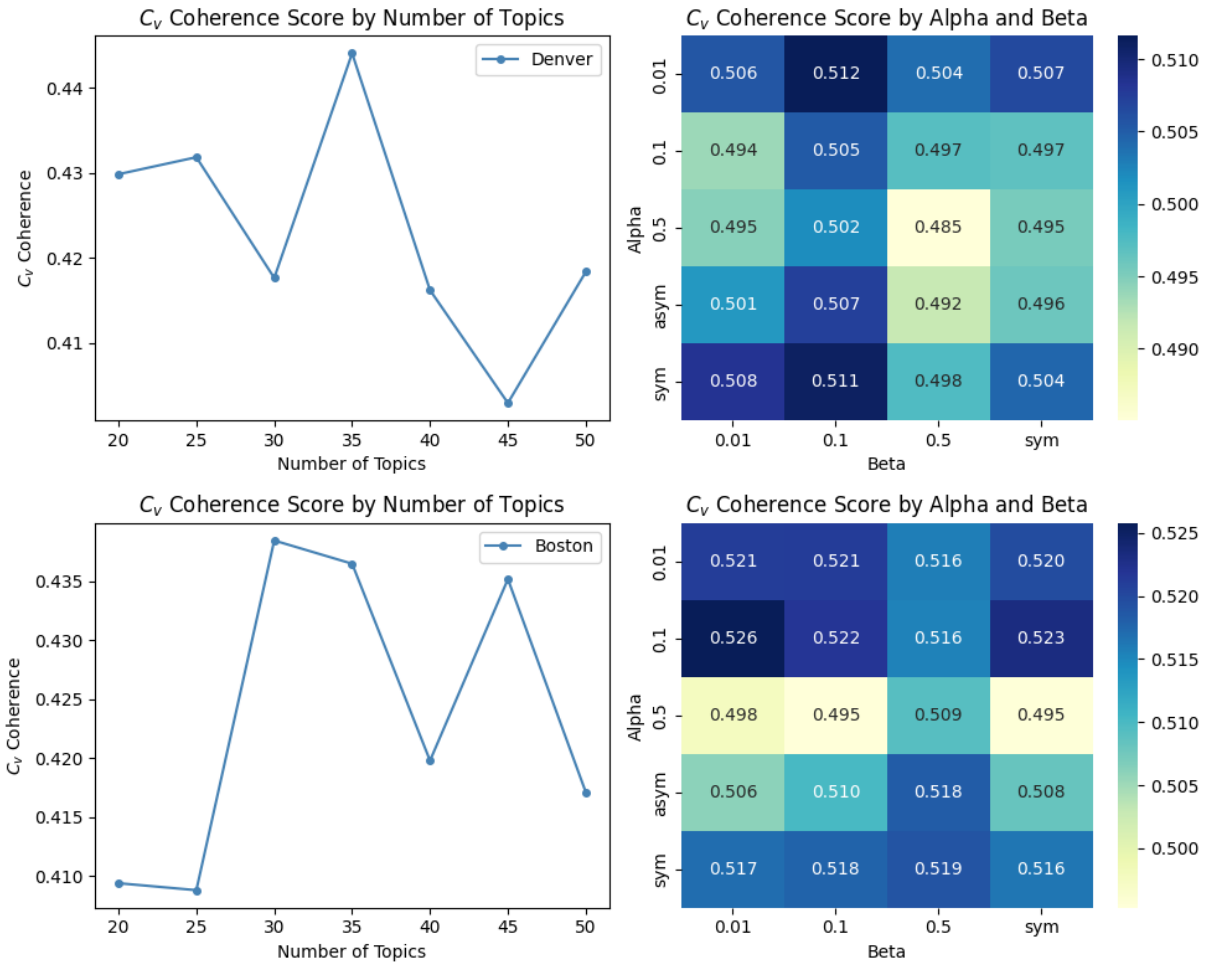


Figure 1: LDA Hyperparameter Tuning

After determining an optimal LDA model, we used pyLDAvis to visualize each topic and its keywords. PyLDAvis has a built-in relevance metric, lambda, which can be scaled to control the specificity of keywords associated with each topic. A lambda of 1 ranks terms only on their frequency within a topic (i.e. the length of their red bars in Figure 2.), while a lambda of 0 ranks terms based on the ratio of their frequency within a given topic to their overall corpus-wide term frequency (i.e. the ratio of red bar to blue bar in Figure 2.) The result is that a lambda closer to 0 pinpoints terms used frequently in a given topic and seldom used elsewhere. It was determined via manual inspection of the results that a lambda of 0.3 provided sets of keywords that were the most coherent and the ideal level of specificity for our aims. As such, we extracted the top 10 keywords for each topic with the lambda value set to 0.3. A pyLDAvis visualization for the Denver data is shown in Figure 2.

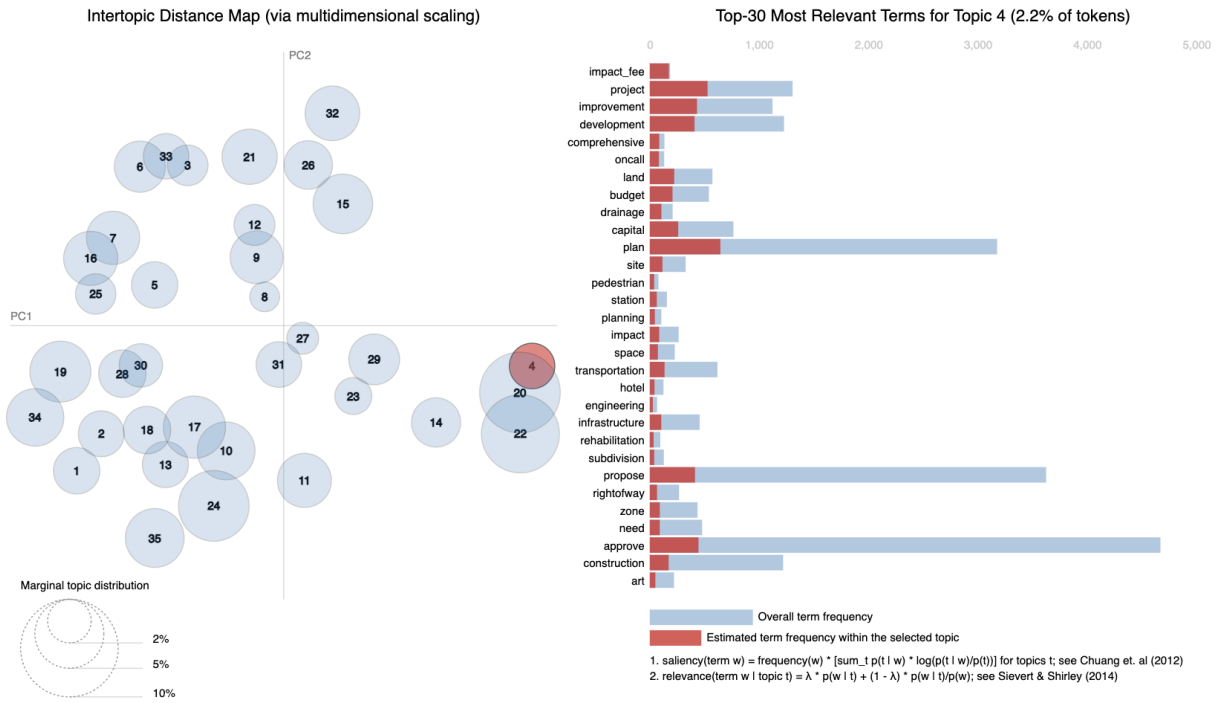


Figure 2: Denver LDA Topic Visualization

Once we extracted topics and keywords from both the Denver and Boston documents, we fed each set of keywords to OpenAI's GPT-4 model using the prompt "Give me a one or two word 'label' that corresponds to these keywords". Using a combination of the results from GPT-4, and our own manual labeling, we arrived at a set of topic-phrases which best encompassed each LDA topic. For example, we labeled the category displayed in the visualization in Figure 2, containing keywords "impact_fee", "project", "improvement", "development", "comprehensive", "oncall", "land", "budget", "drainage", "capital" as "Planning & Urban Development". After associating each set of keywords with a topic-phrase, we manually selected 12 topic-phrases for each city that we thought were most likely to be discussed on Reddit. We then queried Reddit for posts using the labeled topic-phrases as query terms. The topic-phrases selected for querying are displayed in Table 1.

Denver	**Boston**
Planning & Urban Development	Architectural Design
Law Enforcement & Safety	Cannabis
Events & Culture	Parking
Governance & Finance	Public Transportation
Education	Zoning
Infrastructure	Plants & Landscaping
Violations & Penalties	Shoreline & Water
Zoning & Housing	Mixed Use Housing
Justice & Social Services	Signage & Advertising
Property & Real Estate	Residential Types & Living Arrangements
Vehicles	Building Design & Storefronts
Animal Control	Population Density

Table 1: List of Topic-Phrases for Denver and Boston

Sentiment Analysis

In the sentiment analysis portion of the project, we first combined the title and the main content of each Reddit post to create a combined variable with the relevant text. We then analyzed the tone of each of these combined posts, classifying them as "positive", "negative" or "neutral". To achieve this, we used VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon and rule-based sentiment analysis tool, specifically implementing the vaderSentiment library in Python. We selected VADER for this task since it is specifically designed for texts that contain informal language, such as social media posts.

VADER determines sentiment by evaluating the polarity of individual words in a text, considering not only their "emotional intensity" via predefined sentiment scores but also contextual modifiers such as punctuation, capitalization, and intensifiers. It then aggregates these scores to determine an overall sentiment rating for the whole text, represented by a weighted "compound score" between –1 (extreme negative) and +1 (extreme positive) (GeeksforGeeks). VADER also returns the individual sentiment percentages for each text. These scores allow us to assess not only the sentiment classification but also the intensity of the sentiment.

To classify posts as negative, neutral, or positive, we followed the guidance in the library's documentation and started by using –0.05 and +0.05 as the thresholds for the compound score. For example, in Table 2, the Reddit post was rated as 0.0% negative, 85.3% neutral, and 14.7% positive. The compound score is 0.7506, so the overall rating is positive because this value is greater than 0.05. In other words, it is more positive than negative. We also tested additional threshold values for the compound score (±0.2, ±0.5, and ±0.8) to see how they would impact the results. Using larger values means the sentiment must be considered more "extremely" positive, for example, to be predicted as positive.

Municipality	**Denver**	**Boston**
Topic	Infrastructure	Public Transportation
Post Content	Where do I go to report positive feedback for our city's infrastructure? I see so many bike lanes and other ped-friendly infrastructure going up these days each week when I take my little guy out for walks. I just wanna make sure my voice is heard in how happy I am to see these changes! The only thing I'd change is I wish we'd get these done faster and do more of them! Denver can be such a great biking city - I hope to keep seeing these improvements in full force!	Logan public transport: a bad joke. Honestly, how is Boston still so completely backwards with public transit from the airport? The Silver Line (man is that an aspirational bit of branding) is the worlds slowest bus operating on a demented persons timetable (why does it stop at courthouse for 15 minutes?!?). It took us 45 minutes to go from Logan to South Station to get on a commuter rail that runs once every 2 hours. Is this the 1980's?
Sentiment Dictionary	`{'neg': 0.0, 'neu': 0.793, 'pos': 0.207, 'compound': 0.9701}`	`{'neg': 0.084, 'neu': 0.916, 'pos': 0.0, 'compound': -0.7877}`
Overall Rating	Positive	Negative

Table 2: Sentiment Analysis Examples

After calculating the sentiment for each Reddit post, we aggregated results by topic for each city to determine whether certain topics are generally associated with more positive or negative posts. In Denver, posts about "Events & Culture" were the most positive, whereas posts about "Violations & Penalties" were the most negative. Posts pertaining to "Planning & Urban Development" and "Justice & Social Services" also tended to be positive in Denver, indicating that constituents felt the city is performing well in these areas. Topics like "Vehicles" and "Zoning & Housing" had higher proportions of negative posts, indicating that residents were dissatisfied with the current state of these areas. This outcome suggests

potential opportunities for increased attention and legislation in Denver pertaining to civil penalties, vehicles, and housing.

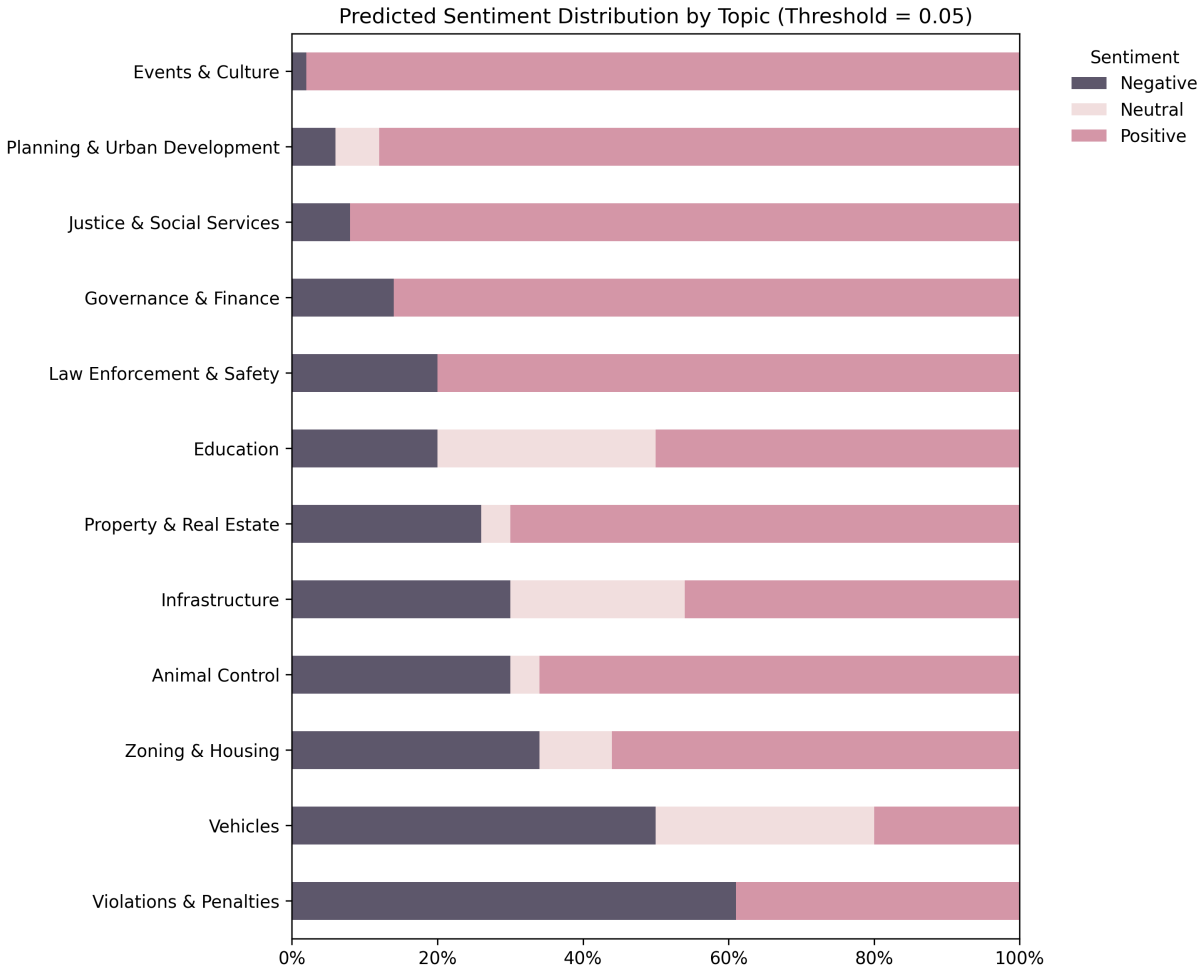


Figure 3: Denver Aggregated Sentiment Distribution

In Boston, posts about "Plants & Landscaping" were entirely positive, while the posts on "Cannabis" were the most negative, closely followed by "Parking" and "Public Transportation". This outcome indicates that cannabis, parking, and public transportation may present opportunities for Boston policymakers to reassess current municipal legislation or consider introducing additional legislation. Interestingly, posts about "Population Density" as well as "Zoning" garnered large proportions of neutral posts in Boston, indicating that these topics may not be areas of strong concern for residents. Overall, it appears that Denver had a larger proportion of negative posts across all topics, as well as fewer posts classified as neutral than Boston, suggesting that Denver residents expressed more dissatisfaction as well as more polarized opinions than Boston residents.

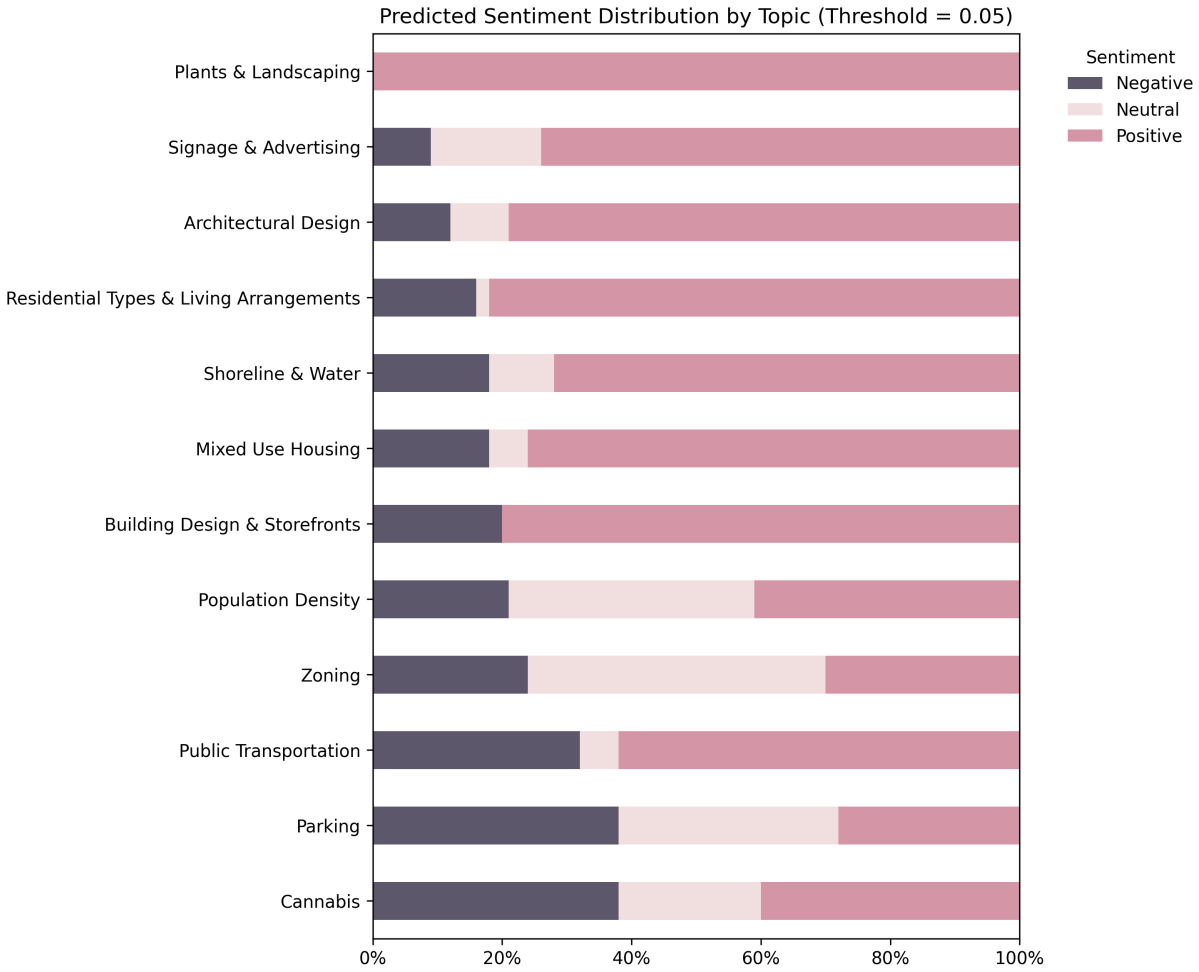


Figure 4: Boston Aggregated Sentiment Distribution

Evaluation

To evaluate the effectiveness of the LDA topic model in categorizing municipal ordinances into distinct topic areas, we used a combination of coherence scores and manual validation. Coherence measures the degree of semantic similarity between keywords in each topic by analyzing their co-occurrence patterns. We calculated two coherence scores using the Gensim library‘s built-in methods: C_v coherence and U_{mass} coherence. The C_v coherence applies a sliding window approach to capture co-occurrence patterns before using normalized pointwise mutual information (NPMI) to quantify word association. Cosine similarity is then used to measure the overall topic consistency based on the word associations. Alternatively, U_{mass} coherence counts co-occurrences directly from the training documents and calculates the log-likelihood of word co-occurrence as a measure of topic coherence (Hui). Topic coherences can be averaged over all topics in a model to determine a model coherence score which reflects the performance of the overall LDA model. The averaged model coherence scores for both coherence metrics are displayed in Table 3.

Coherence Score	**Denver**	**Boston**
C_v	0.512	0.526
U_{mass}	-2.030	-2.003

Table 3: Coherence Scores for Denver and Boston

Overall, our LDA models for both cities produced decent coherence scores. C_v coherence scores fall between 0 and 1, so scores above 0.5 indicate reasonable coherence and are generally acceptable. Therefore, the scores achieved by our models are considered satisfactory. For U_{mass} coherence scores, less negative values are better, with scores as low as -14 being possible (StackOverflow). As such, our U_{mass} coherence scores are also within an acceptable range, indicating decent model performance. It is worth noting that U_{mass} coherence tends to decrease as the number of topics increases, meaning that our relatively large number of topics may have contributed to the observed values.

In addition to using coherence scores, we manually evaluated the model's ability to produce well-defined topics with clear and relevant keywords. We assessed whether the identified topics captured meaningful subject areas present in the municipal ordinances and ensured that no significant topics were overlooked. The model that best met these criteria was selected, and based on our manual review of the LDA model's output, we concluded that our final choice produced relevant, meaningful, and coherent topics.

To evaluate the VADER sentiment analysis, we manually labeled a sample of posts in our dataset. We selected five posts per topic for each city, resulting in 60 Reddit posts for Denver and 60 posts for Boston. We attempted to create a relatively balanced sample by aiming to include at least one post in each sentiment category for each topic. By comparing this manually labeled dataset to VADER's predicted classifications, we calculated accuracy, precision, recall, and F1 score to assess the model's performance. We also created a 3×3 confusion matrix to analyze common misclassifications and determine whether the model struggles to differentiate between certain sentiment classes.

Table 4 presents the accuracy, precision, recall, and F1 score for various threshold values of the compound score. The Denver model demonstrated moderate performance under the baseline threshold of 0.05, with an accuracy of 0.52. The Boston model outperformed Denver with a higher accuracy of 0.65. Both models exhibited similar precision and recall, though Boston's recall slightly surpassed Denver's. The F1 scores for both models were also reasonable. Overall, sentiment analysis on both datasets yielded acceptable performance.

Threshold	**Accuracy**	**Precision**	**Recall**	**F1 Score**
±0.05	0.52	0.69	0.53	0.47
±0.2	0.52	0.62	0.53	0.49
±0.5	0.53	0.58	0.56	0.52
±0.8	0.50	0.57	0.52	0.50

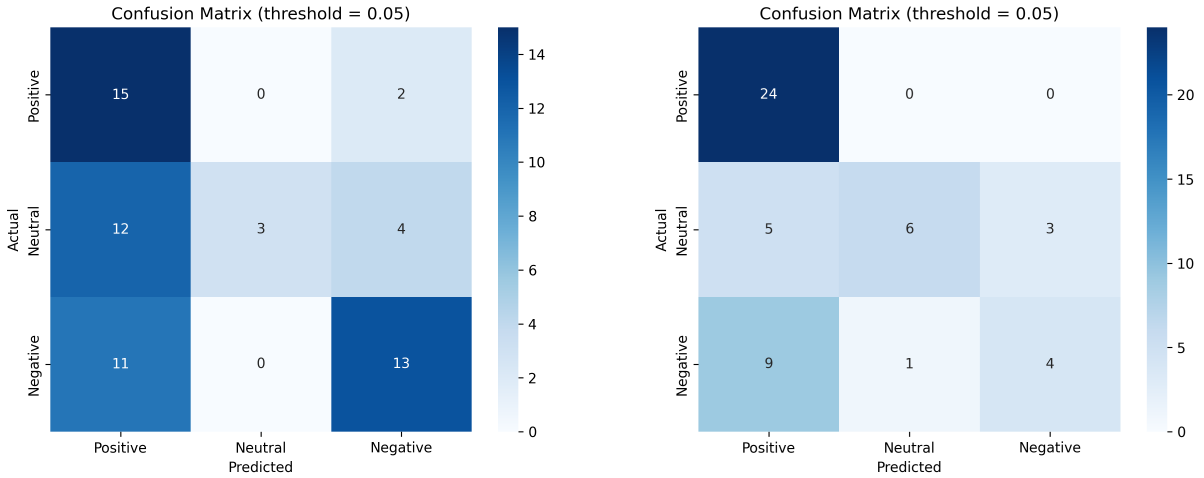
Table 4: Denver Sentiment Analysis Evaluation

Threshold	**Accuracy**	**Precision**	**Recall**	**F1 Score**
0.05	0.65	0.69	0.57	0.58
0.2	0.65	0.66	0.58	0.59
0.5	0.54	0.54	0.50	0.49
0.8	0.48	0.66	0.47	0.40

Table 5: Boston Sentiment Analysis Evaluation

The confusion matrices for Denver and Boston, using the VADER recommended compound score threshold of 0.05, are shown in Figure 5. The confusion matrix for Denver shows that a significant number of posts which are actually neutral or negative are predicted to be positive, indicating that the model likely has a bias towards classifying text as positive. In comparison, there are relatively few other misclassifications in the Denver dataset, with only 2 negative posts classified as neutral and 4 negative posts classified as neutral.

The confusion matrix for Boston shows a more balanced performance, though still with a disproportionate number of posts predicted as positive. The most common misclassification error was negative posts predicted as positive, followed by neutral posts predicted as positive. On the other hand, there are no posts that are actually positive but classified as negative, which suggests that the model also exhibits a bias toward classifying text as positive on the Boston dataset, albeit a more subtle one.



(a) Denver

(b) Boston

Figure 5: Confusion Matrices for Denver and Boston

Limitations

Our methodology introduces a technique for comparing municipal legislation to community opinion, although it does present a number of limitations in its topic modeling, sentiment analysis, and overall approaches.

In terms of topic modeling, the main limitations are the difficulty in topic interpretibility, and relatedly, the necessity of manual intervention to determine an ideal lambda value and encompassing topic-phrases. LDA, through its probabilistic nature, determines the words most likely to be assigned to each topic, but this approach does not always correspond to the most easily interpretable, cohesive topics since factors like word order or semantic meaning are not considered. As a result, words such as "ongoing", "deny", or "impact_fee", which may show up often in a topic but which do not convey significant meaning, may be selected as representative topic keywords and obscure the overall topic theme. A potential solution to this issue could involve using a more advanced embedding model that incorporates semantic meaning before feeding the documents to LDA.

Manual validation was used to choose a lambda value which promoted topic interpretibility, as well as select topic-phrases which translated sets of keywords, not all of which were informative, into cohesive themes. While this validation helped mitigate the difficulty in topic interpretability, it is time consuming and not scalable. The need for manual validation may also be alleviated by semantic-aware embeddings, but future methods could additionally introduce a more automated strategy for generating topic-phrases.

Limitations to sentiment analysis include the nuanced nature of Reddit posts and the positive bias exhibited by the VADER model on our datasets. Posts may express both positive and negative opinions on different matters, use sarcasm, or include lists, suggestions, and reports which are difficult to pinpoint as positive or negative in sentiment. Additionally, our results indicated that the VADER sentiment analysis model tended to classify neutral and even negative posts as positive. We were not able to mitigate this issue by increasing the compound score threshold for positive classification, as this decreased the accuracy

of our model by causing it to make different misclassifications. These issues provide a barrier to quantifying and understanding the true nature of public opinion, although they could potentially be mitigated by using more advanced NLP models, or through incorporation of additional post metadata, such as comments and engagement, into our methodology.

Finally, challenges to our overall approach include the limited sample size of relevant Reddit posts on each topic, and the question of next steps for policymakers given our results. During our search, we found that some topic searches returned off-topic posts, and in some cases, we were unable to meet the threshold of 50 posts per topic. This issue could potentially be mitigated by using a different search algorithm, aside from the Reddit-internal one, or by introducing a method for filtering out off-topic posts. On the topic of data sparsity, it is also important to note that topics on which there is limited municipal legislation are not considered in this analysis since we extract topics from municipal data, although these topics may be of interest to residents and consequence to legislators.

Regarding next steps, while the results of our research point legislators toward a specific topic where residents express dissatisfaction, they do not provide insight into the underlying causes of that dissatisfaction or which specific issues need to be addressed. This makes it difficult for policymakers to translate these results into actionable change. Realistically, it may be more useful for legislators to be shown the most relevant or highest engagement posts for each topic, rather than just the sentiments, as this would provide more context and a clearer direction for potential legislative action.

Conclusion

In this project, we developed a novel approach to bridge the gap between local governments and their residents by comparing municipal legislation to community opinion. We utilized a combination of LDA topic modeling and VADER sentiment analysis to extract insights from the municipal ordinances and the Reddit communities of Denver and Boston and concluded that both models achieved acceptable performance. Our analysis revealed Denver residents feel satisfied with the events, culture, and urban development of their city, but are discontented by zoning, housing, vehicles, and civil penalties. In Boston, we found residents pleased with their city's landscaping and architectural design, yet dissatisfied with public transportation, parking, and cannabis laws. These findings offer a road map for city council members and policymakers, highlighting the areas that require greater attention and potential legislative action. By leveraging these insights during legislative sessions, lawmakers in both cities can prioritize changes that will improve the quality of life for their residents.

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